Lab 8
Arrays
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Sample Programs

Program to copy 1 array to another array.

Note: We have used indexed addressing this time to iterate through array.

```
.data
numbers         DW      34, 45, 56, 67, 75, 89
numbers2        DW      6 DUP (0)                   ;initialize numbers2 with 0,0,0,0,0,0 (6 times)

.code
main proc
        mov ecx, 6                          ;initialize counter to array size
        mov esi, 0                          ;initialize offset to 0

move_next:
        mov ax, [numbers + esi]   ;store value at numbers + offset to eax
        mov [numbers3 + esi], ax ;store value in ax to numbers2 + offset
        add esi, 2                          ;increment your offset
        loop move_next              ;loop until ecx == 0

        Invoke ExitProcess, 0

main endp
end main
```

Lab Work

- Take reference from the above sample program and implement following programs.
- Attach properly commented code in the lab report.
- Add flow chart of your logic for each program implemented.

Question 1:

Write a program that finds factorial of given variable n=5 and store it into fact variable.

Question 2:

You have been given an array of numbers and two variables even_sum and odd_sum. Write a program that checks each element in array, if it is even number then it will be added to even_sum and if it is odd number then it will be added to odd_sum.

```
.data
numbers         DW      34,  45,  56,  67,  75, 89
odd_sum         DW      ?
even_sum        DW      ?
```

Hint: You can do TEST AX,01, if it is 0 then jump to add AX to even sum

```
{ TEST AX, 01 ;
JZ add_even ;
ADD BX, AX }
```

(If DX stores even_sum and BX stores odd_sum, of course CX has count)